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Faculty of Computer Science and Engineering
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Real-Time Object Detection for Autonomous Vehicles Using YOLOv8 and Enhanced Hybrid Frameworks

Transfer Learning, Fine-Tuning and Detailed Architectural Analysis on a Custom 13-Class Dataset

A Graduation Project Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Computer Science and Engineering

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UNDERTAKING & DECLARATION

We, the undersigned, hereby declare that this graduation project report titled **"Real-Time Object Detection for Autonomous Vehicles Using YOLOv8 and Enhanced Hybrid Frameworks"** is the result of our own original research and development work conducted at Alamein International University.

We certify that all materials, data, and contributions from other sources or publications have been clearly identified, properly cited, and acknowledged within the bibliography section. This work has not been submitted, either in whole or in part, for any other degree or professional qualification at this or any other academic institution.

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Abstract

Accurate and real-time object detection is a fundamental requirement for autonomous driving systems, where the ability to identify pedestrians, vehicles, traffic signs, and road hazards in milliseconds is a critical component for perception stacks. This graduation project focuses on building a real-time, high-accuracy object detection framework tailored for urban road scenes, particularly in challenging environments such as Cairo, Egypt.

This thesis presents an empirical research framework that evolves through progressive phases to address the core speed-accuracy trade-offs. Initially, a state-of-the-art YOLOv8 medium variant (YOLOv8m) baseline is established. A custom dataset was created by filtering and re-annotating images from the large-scale Microsoft COCO dataset, reducing the original 80 classes to 13 classes highly relevant to autonomous driving. This filtering concentrates the model's learning capacity on domain-specific objects. Two-stage training was employed: first, transfer learning with the backbone frozen for 100 epochs, followed by full fine-tuning for another 100 epochs to allow the entire model to adapt to Egyptian road conditions, lighting variations, and object scales.

To further address single-stage limitations on small and distant objects, the **YOLO++** architecture is introduced, incorporating four targeted enhancements: (1) Squeeze-and-Excitation (SE) attention blocks for channel-wise feature reweighting, (2) neck-only attention placement for improved multi-scale feature fusion, (3) a refined PAN-FPN structure for small object feature preservation, and (4) optimized loss scheduling with cosine annealing. YOLO++ achieves a significant optimization gain with faster inference compared to the baseline.

Finally, a **Hybrid Fusion Framework (YOLO++-RCNN)** is proposed, integrating YOLO++ feature extraction with a Faster R-CNN ResNet50-FPN backbone through an **Adaptive Feature Fusion (AFF)** module. The fusion learns dynamic per-channel attention weights to combine complementary strengths: YOLO++'s speed and recall versus Faster R-CNN's localization precision. Performance is evaluated using standard metrics across six model configurations in comprehensive ablation studies. The results demonstrate the effectiveness of transfer learning, attention mechanisms, and multi-paradigm hybrid fusion for real-world autonomous vehicle perception.

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Chapter 1

Introduction

1.1 Background of Object Detection in Autonomous Vehicles

The emergence of autonomous vehicles (AVs) represents one of the most transformative technological shifts of the 21st century. From advanced driver-assistance systems (ADAS) to fully self-driving platforms, the ability to perceive and interpret the surrounding environment in real time is the cornerstone of vehicular autonomy. At the heart of this perception pipeline lies **object detection** — the task of simultaneously localizing and classifying objects of interest within an image or video frame.

The vehicle must detect and classify other road users (pedestrians, cyclists, vehicles), traffic control devices (lights, signs), and static/dynamic obstacles in real time, under varying conditions: daylight, night, rain, fog, crowded urban scenes, and high-speed highways. Traditional approaches relied on hand-crafted features (HOG, SIFT) combined with sliding-window classifiers or classical machine learning. These methods were limited in accuracy and robustness. The deep learning revolution enabled end-to-end learning of hierarchical features, dramatically improving detection performance.

1.2 Motivation for This Project

In Egypt, particularly in densely populated cities like Cairo, autonomous vehicle perception faces unique challenges:

- Extremely crowded roads with mixed traffic (cars, microbuses, motorcycles, pedestrians, animal crossings).
- Variable lighting conditions (harsh sunlight, night driving, tunnel transitions).
- Poorly maintained infrastructure (faded signs, missing lane markings).
- Small and distant objects (traffic lights, parking meters) that must be detected reliably.

Pre-trained models on COCO perform well on general objects but need domain adaptation for these conditions. Furthermore, contemporary single-stage detectors offer exceptional inference speed but can exhibit limitations in small object localization and precise boundary

regression. Two-stage detectors provide superior localization accuracy through their region proposal mechanism, but at the cost of significantly reduced inference speed. This project addresses this gap by establishing a strong baseline, enhancing it through architectural modifications (**YOLO++**), and ultimately proposing a novel hybrid detection framework that combines the complementary strengths of two different detector families.

1.3 Problem Statement

The central research problem addressed in this thesis is: *Can an optimized architecture or a hybrid framework be designed that intelligently fuses multi-paradigm feature representations, thereby achieving detection quality that surpasses standalone single-stage architectures in isolation on small and rare objects, while maintaining inference speeds appropriate for real-world deployment?*

1.4 Objectives

The project pursues the following primary objectives:

1. Create and preprocess a custom 13-class dataset from COCO annotations tailored for road-relevant categories.
2. Implement and compare historical object detection models with modern one-stage detectors.
3. Provide an exhaustive architectural breakdown of the YOLOv8 medium variant baseline.
4. Apply two-stage training: transfer learning (frozen backbone) followed by full fine-tuning.
5. Design, validate, and implement **YOLO++**, incorporating Squeeze-and-Excitation (SE) attention, neck-focused feature refinement, and optimized loss scheduling.
6. Propose and train the **YOLO++-RCNN Hybrid Adaptive Fusion Framework**, combining YOLO++ and Faster R-CNN feature extractors through a learnable attention-gated fusion module.
7. Conduct comprehensive ablation studies across six model configurations to quantify the contribution of each component.

1.5 Contributions of This Work

- A domain-specific 13-class dataset filtered and optimized for urban road scenes.
- In-depth explanation and evaluation of every block in YOLOv8m (Conv, C2f, SPPF).
- Development of the enhanced **YOLO++** architecture via neck-only attention placement and refined PAN-FPN multi-scale structures.

- Implementation of an **Adaptive Feature Fusion (AFF)** module providing a complete end-to-end trainable hybrid detection pipeline.
- Comprehensive six-model ablation study quantifying the precision, recall, mAP, and FPS trade-offs.

1.6 Thesis Structure

Chapter 2 reviews historical models and related work. Chapter 3 explains the baseline YOLOv8 architecture in technical depth. Chapter 4 details custom dataset creation and preprocessing. Chapter 5 describes the methodology of transfer learning and fine-tuning. Chapter 6 introduces the engineered YOLO++ modifications and the proposed Hybrid Fusion Framework. Chapter 7 presents the experimental results and comprehensive ablation studies, followed by a critical discussion and conclusions in Chapter 8.

Chapter 2

Related Work and Historical Object Detection Models

2.1 Early Approaches: Deformable Parts Model (DPM)

The Deformable Parts Model (Felzenszwalb et al., 2010) was a landmark before deep learning. It used Histogram of Oriented Gradients (HOG) features, a root filter for the whole object, and part filters with deformation costs. Detection was performed via sliding window and dynamic programming. DPM suffered from low accuracy on complex scenes, very slow inference ($\sim$1 FPS), and poor generalization to scale or rotation changes.

2.2 R-CNN (Regions with CNN features) – Girshick et al., 2014

R-CNN marked the beginning of deep-learning-based detection:

1. Generate $\sim$2000 region proposals using Selective Search.
2. Warp each region to a fixed size (227×227).
3. Feed into a pre-trained CNN (AlexNet) to extract features.
4. Classify with SVM and refine via bounding-box regression.

Accuracy was high ($\sim$50–60% mAP on PASCAL VOC), but inference was extremely slow ($\sim$47 seconds per image) due to independent forward passes for each proposal.

2.3 Fast R-CNN – Girshick, 2015

Fast R-CNN introduced major structural improvements:

- Executing a single CNN forward pass on the entire image rather than proposals.
- Introducing an Region of Interest (RoI) pooling layer to extract fixed-size features from the global feature map.

- Multi-task loss combining classification and regression branches.

Speed improved significantly to ∼0.2 seconds per image, with an accuracy of ∼65–70%.

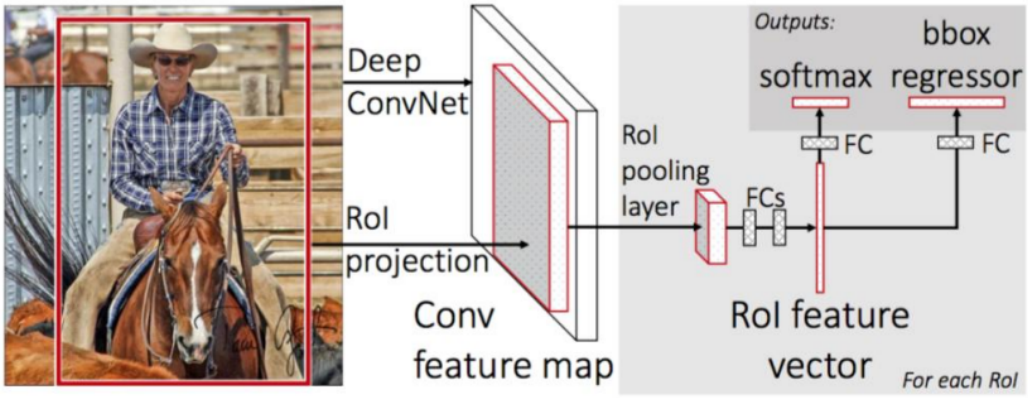


Figure 2.1: Fast R-CNN architecture with RoI pooling

2.4 Faster R-CNN – Ren et al., 2015

Faster R-CNN introduced the Region Proposal Network (RPN), replacing selective search completely:

- A shared convolutional backbone extracts global image feature maps.
- The RPN predicts objectness scores and bounding box deltas at multiple scales and aspect ratios using anchor boxes.
- The entire pipeline became fully end-to-end trainable.

Inference speed reached 5–17 FPS, with an mAP of ∼70–75% on COCO. It remains a standard benchmark for localization accuracy, particularly in scenarios requiring precise boundary regression.

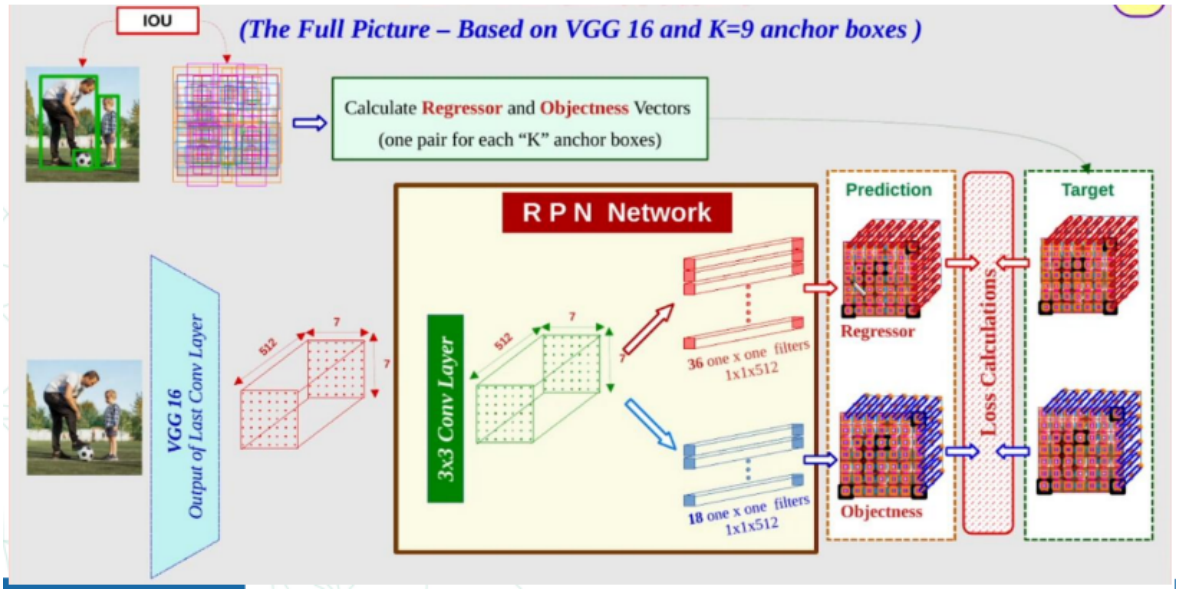


Figure 2.2: Faster R-CNN with RPN

2.5 Comparison Table – Two-Stage vs One-Stage Detectors

Table 2.1: Detailed comparison of detection paradigms

Model	Type	Speed (FPS)	mAP (COCO)	Parameters	Key Strength	Key Weakness
DPM	Classical	∼1	Low	Very low	Interpretable	Poor accuracy
R-CNN	Two-stage	∼0.02	50–60%	Medium	High accuracy	Extremely slow
Fast R-CNN	Two-stage	0.5–5	65–70%	Medium	Faster training	Proposal bottleneck
Faster R-CNN	Two-stage	5–17	70–75%	∼41.8M	End-to-end	Slower than one-stage
YOLOv1 (2015)	One-stage	45	∼60%	Low	Real-time	Lower accuracy
YOLOv3	One-stage	30–60	∼65%	Medium	Good balance	Anchor-based
YOLOv8m	One-stage	80–120	Custom ∼75%	∼25.9M	Fast + accurate	Small-object challenge

Chapter 3

YOLOv8 Architecture – Detailed Technical Explanation

3.1 Overview of YOLO Evolution

YOLO (You Only Look Once) was introduced by Redmon et al. (2016) as the first true one-stage detector. Subsequent versions improved anchors, backbone architectures, and loss functions. Ultralytics YOLOv8 (2023) introduced key architectural enhancements: anchor-free detection, a decoupled head, C2f blocks (improved Cross Stage Partial structure), and Distribution Focal Loss (DFL) for bounding box regression. The medium variant (yolov8m.pt) has ∼25.9 million parameters and provides an excellent speed-accuracy baseline.

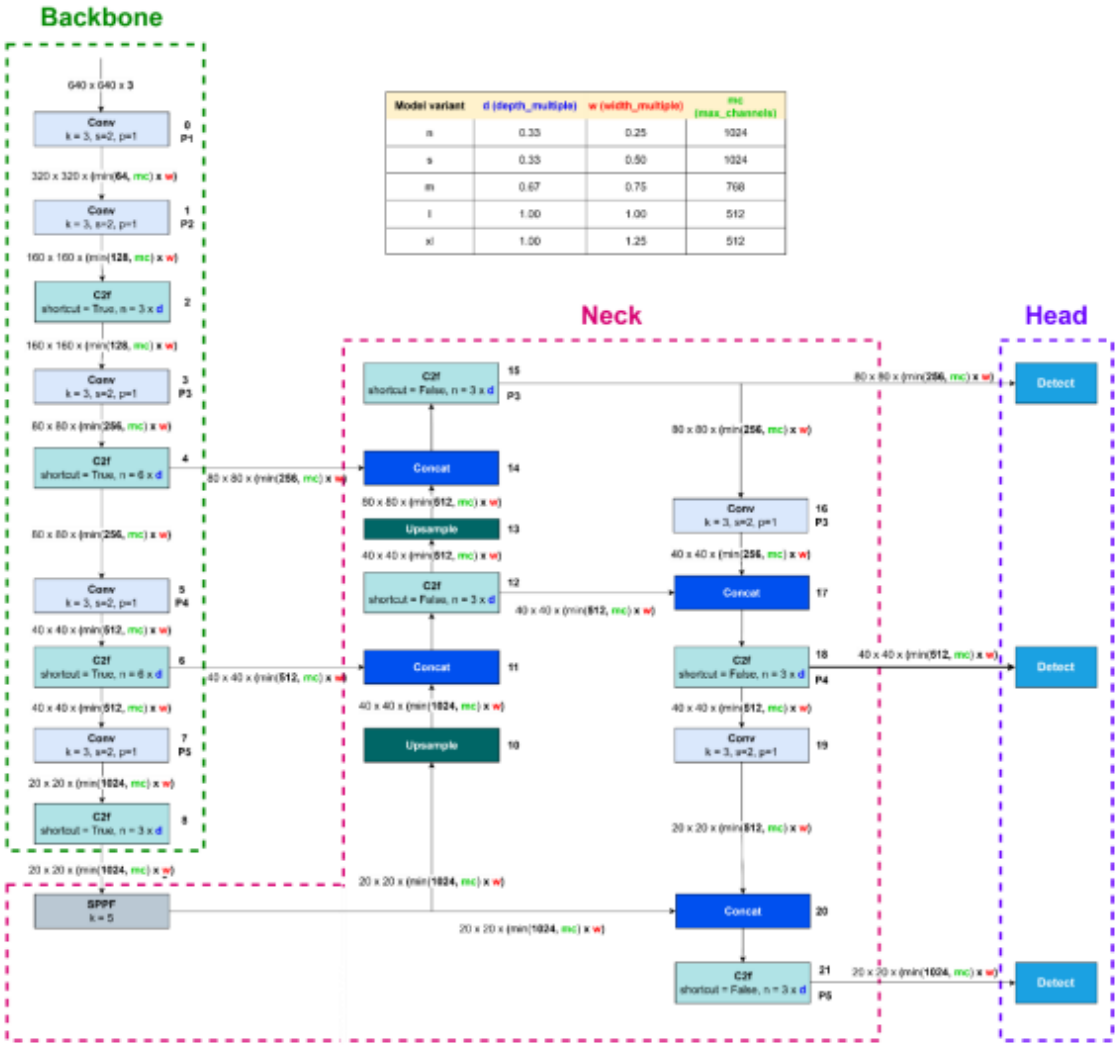


Figure 1: YOLOv8 Architecture (Adapted from [11], [12])

Figure 3.1: Full YOLOv8 pipeline

3.2 Backbone – CSPDarknet with C2f Blocks

The backbone extracts hierarchical features through downsampling and feature refinement.

3.2.1 Initial Conv Blocks (CBS)

Every downsampling step starts with a standard convolutional block containing a 2D Convolution, Batch Normalization, and Sigmoid Linear Unit (SiLU) activation function:

$$\text{Conv2d} \rightarrow \text{BatchNorm2d} \rightarrow \text{SiLU}(x) = x \cdot \sigma(x)$$

SiLU (Swish) is used instead of ReLU because it is smooth and allows small negative gradients to flow, preventing dead neurons.

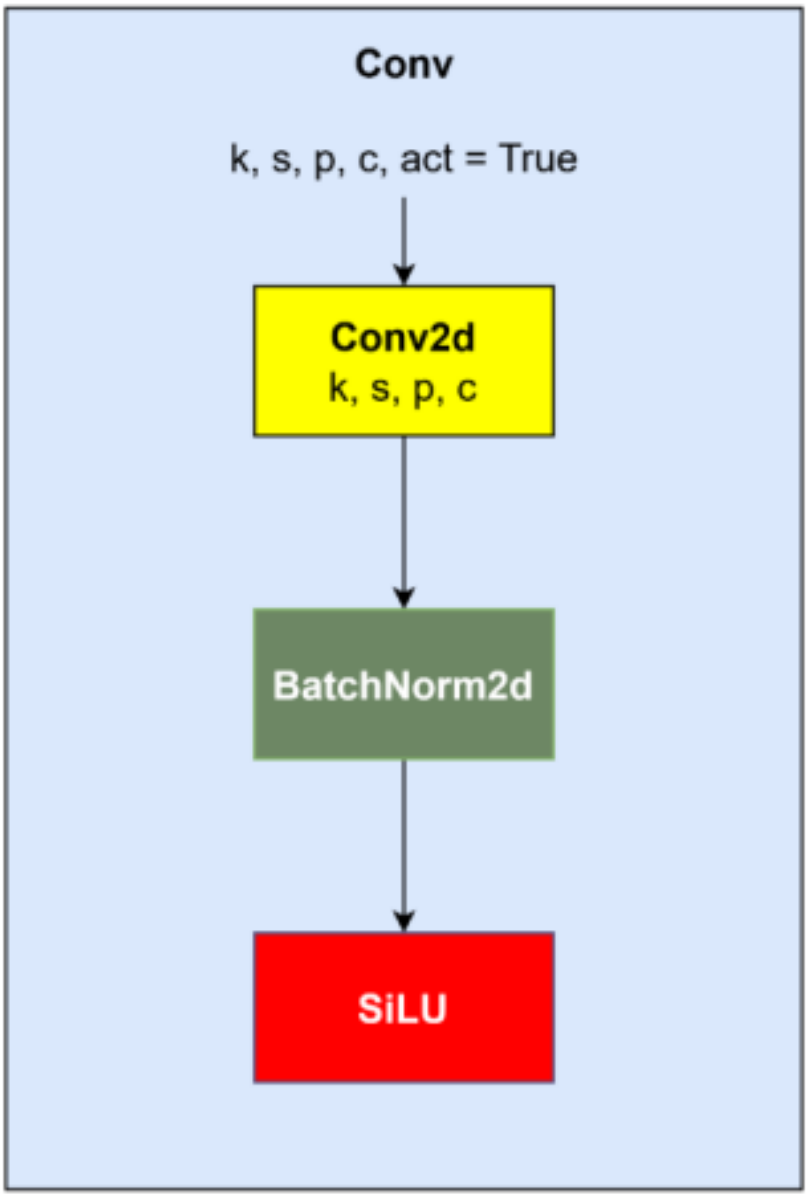


Figure 3.2: Conv → BN → SiLU block

3.2.2 C2f Module – Core Building Block

The C2f (CSP with 2 branches fused) replaces older C3 blocks. It splits channels, processes one half with n bottleneck sub-blocks, keeps the other half as identity, and then concatenates all branch outputs to enrich the gradient flow.

The internal bottleneck structure with a residual shortcut is defined as:

$$\text{Output} = \text{Input} + \text{Conv}_{3\times3}(\text{Conv}_{3\times3}(\text{Input}))$$

Mathematical formulation: Let the input tensor be $X \in \mathbb{R}^{H\times W\times C}$

1. First 1×1 Convolution: $X' = \text{Conv}_{1\times1}(X)$
2. Channel Splitting: $X_1, X_2 \in \mathbb{R}^{H\times W\times C/2}$

3. Chained Bottlenecks on X_2: $Y = \text{Bottleneck}_n(X_2)$

4. Concatenation: $Z = \text{Concat}(X_1, Y_1, Y_2, \ldots, Y_n)$

5. Final 1×1 Convolution: $\text{Output} = \text{Conv}_{1\times1}(Z)$

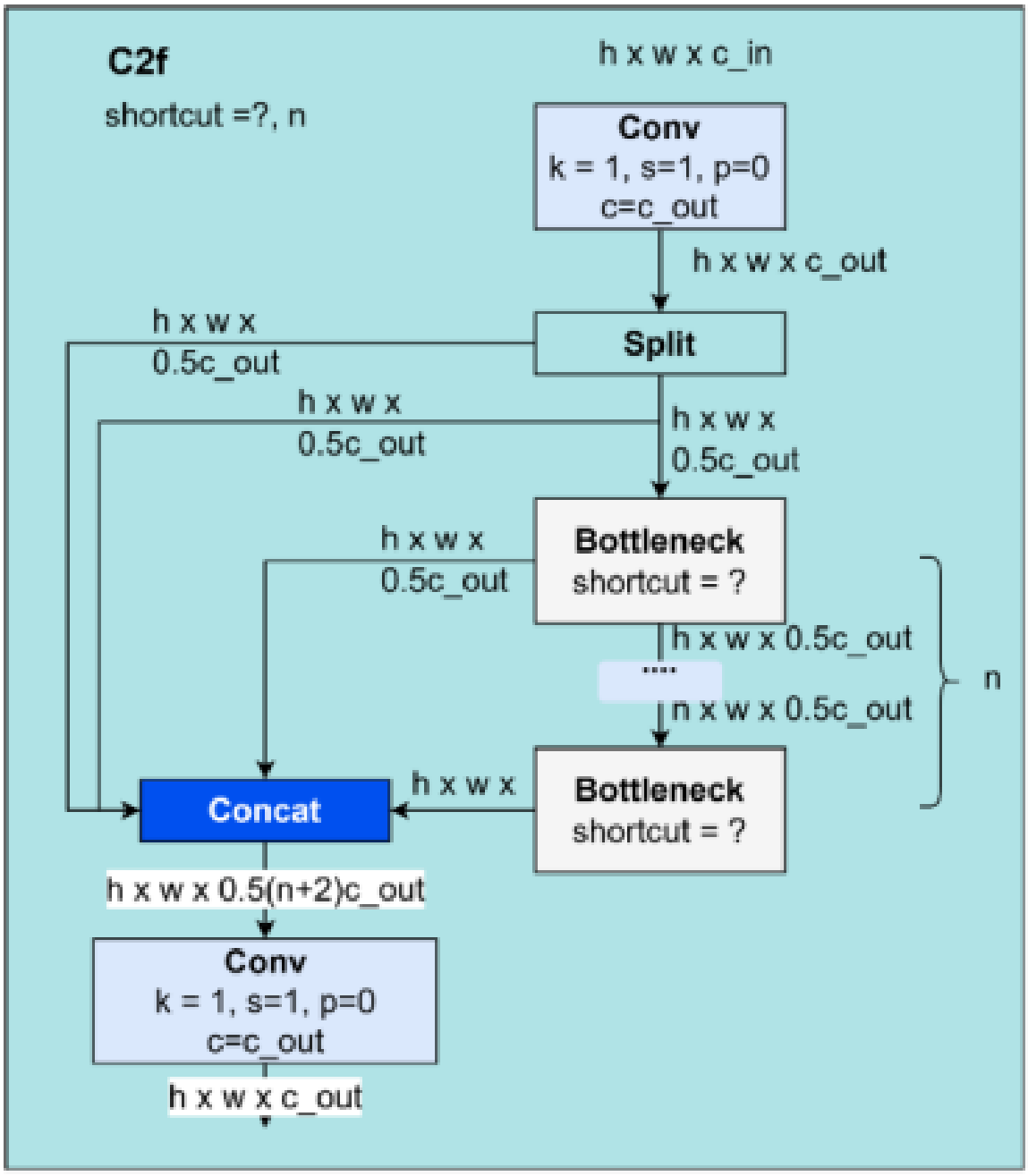


Figure 3.3: C2f block with n bottlenecks

In YOLOv8m, the block depth parameters are scaled such that after the first downsample, C2f has $n = 3$; after the second, $n = 6$; after the third, $n = 6$; and right before the SPPF block, $n = 3$.

3.2.3 SPPF – Spatial Pyramid Pooling Fast

The SPPF module provides multi-scale contextual information efficiently:

1. Initial 1×1 Convolution.
2. Three chained 5×5 Max-Pooling layers (with stride=1, padding preserved).
3. Concatenating the original input along with the three pooled features.
4. A final 1×1 Convolution to compress the expanded channel dimensions.

This sequential pooling configuration achieves identical receptive fields to parallel pooling paths while dramatically lowering GFLOPs.

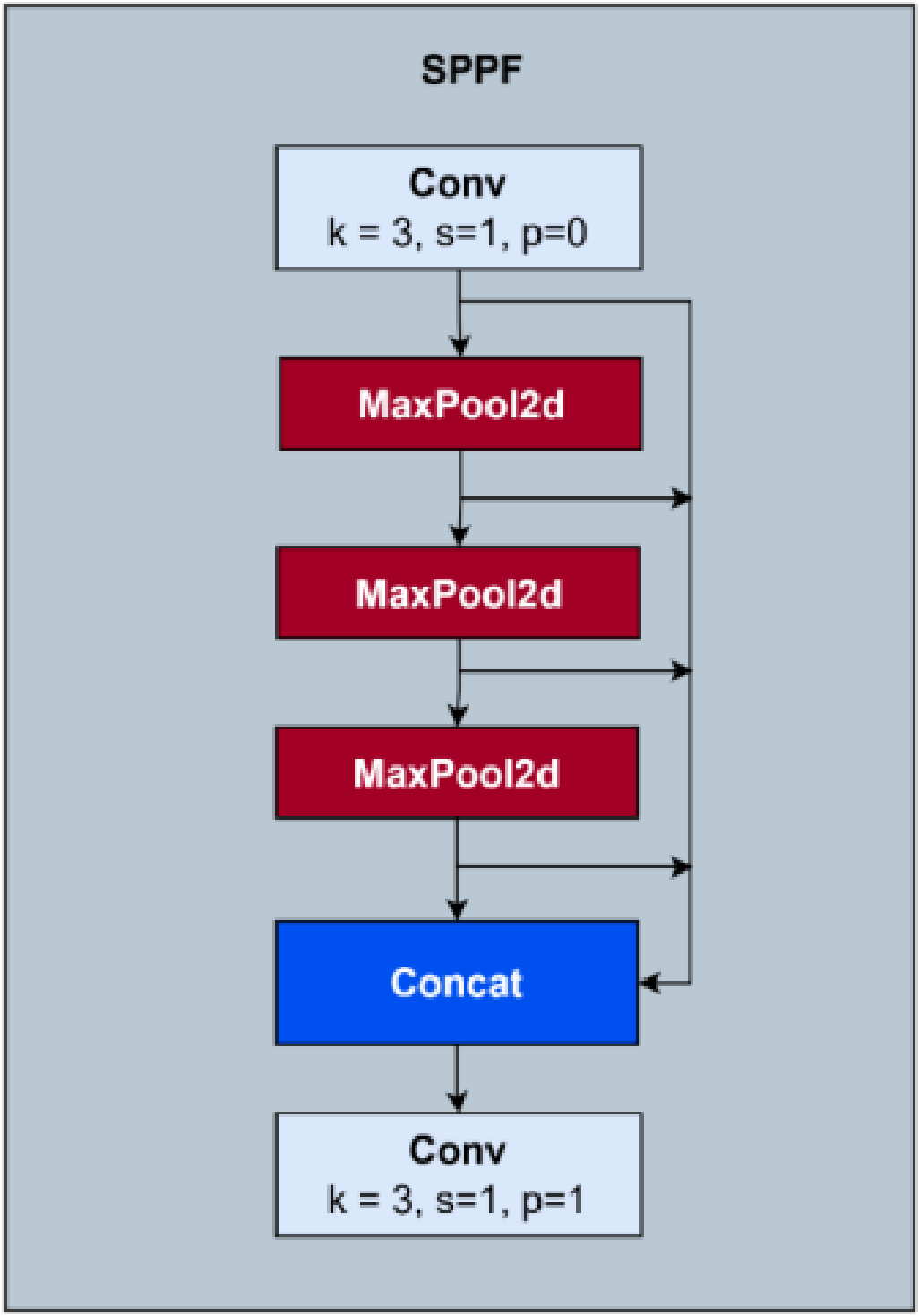


Figure 3.4: SPPF module – chained pooling

3.3 Neck – Feature Pyramid Network + Path Aggregation Network

The neck fuses multi-scale features across different layers of the backbone:

- **Top-down path** (FPN): Deep semantic features are propagated down to high-resolution

maps via upsampling and concatenation with shallow backbone features.

- **Bottom-up path** (PAN): Precise spatial details from high-resolution maps are routed back up to the deep layers via downsampling convolutions and concatenation.

This bidirectional routing creates high-quality feature representations at P3 (80×80), P4 (40×40), and P5 (20×20).

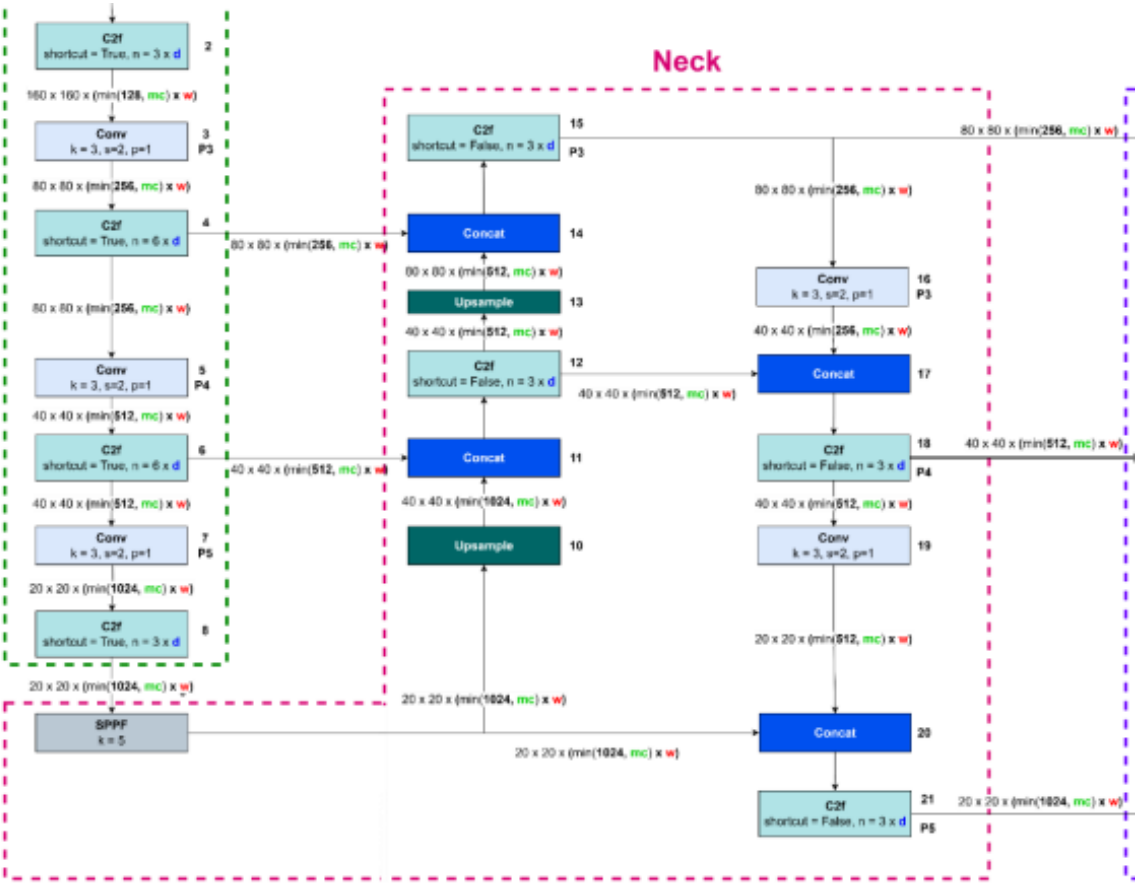


Figure 3.5: Bidirectional neck structure

3.4 Head – Decoupled Anchor-Free Detection

The YOLOv8 head implements a decoupled, anchor-free detection strategy. Classification and regression tasks are separated into independent convolution branches.

The classification branch passes the features through two 3×3 CBS blocks followed by a 1×1 Convolution mapped to $nc = 13$ channels with a sigmoid activation function for object probabilities. The regression branch channels features through two 3×3 CBS layers into a 1×1 Convolution outputting $4 \times \text{reg_max}$ (where $\text{reg_max} = 16$) channels.

Bounding box coordinates are decoded dynamically using **Distribution Focal Loss (DFL)**, which treats box distances as continuous distributions:

$$\text{DFL}(p_i, p_{i+1}) = -\left[y_i \log(p_i) + (1 - y_i) \log(p_{i+1})\right]$$

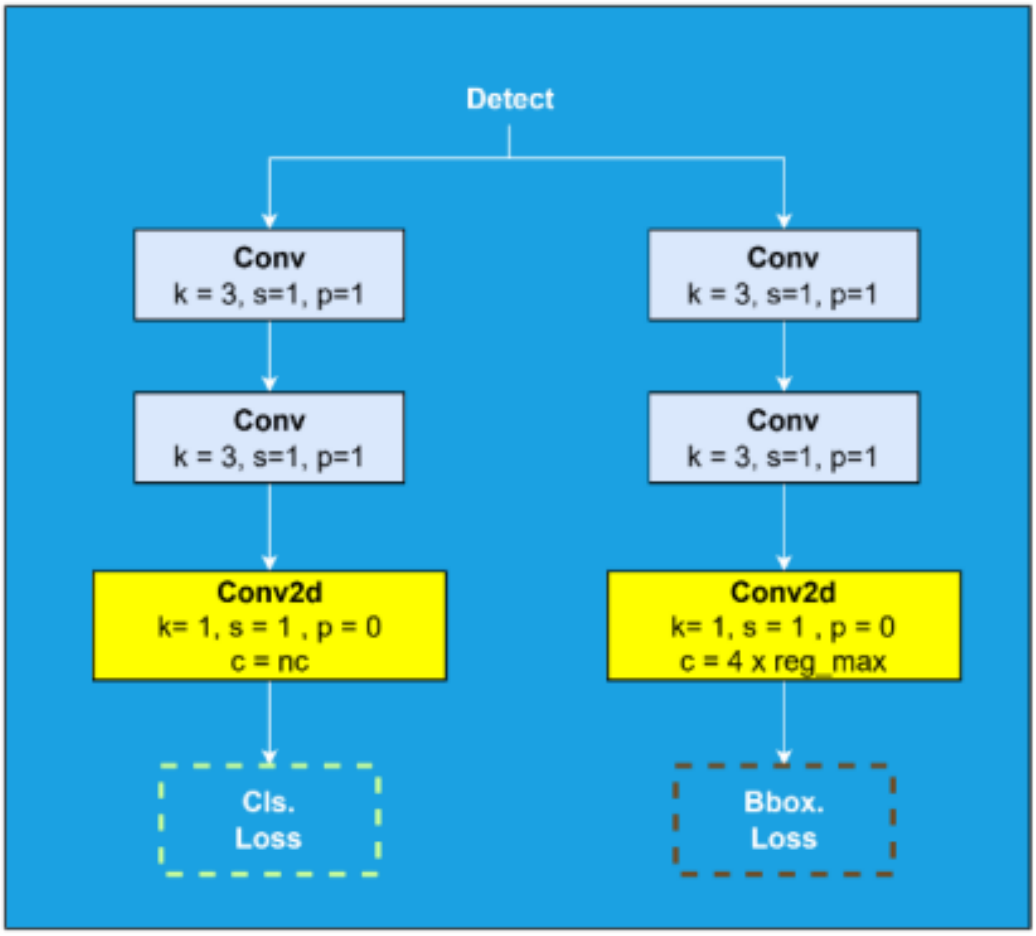


Figure 3.6: Decoupled classification and regression head

3.5 CNN Feature Hierarchy – Edges, Textures, Shapes

Feature representation changes dynamically across layers:

- **Early Layers**: Detect low-level primitive features such as horizontal, vertical, and diagonal edges, corners, and local pixel gradients.
- **Middle Layers**: Assemble primitives into textures and complex repeating patterns.
- **Deeper Layers**: Extract semantic shapes and domain-specific object parts (e.g., wheels, car fronts, pedestrian torsos).

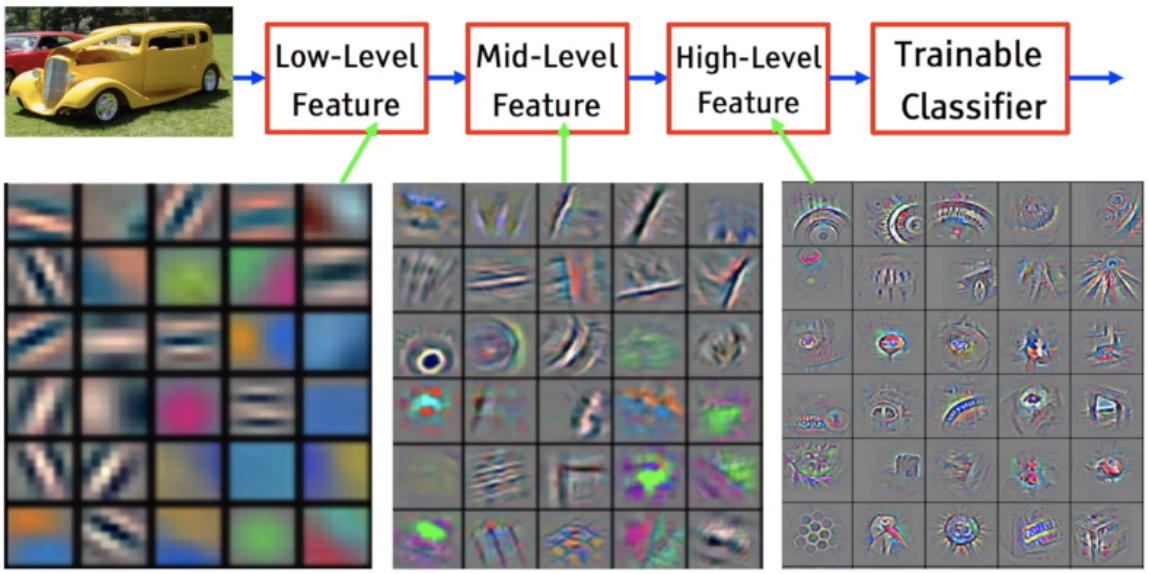


Figure 3.7: Progressive feature complexity in CNN

Chapter 4

Dataset Creation and Preprocessing

4.1 Overview of Microsoft COCO Dataset

The Microsoft Common Objects in Context (COCO) dataset is a widely utilized benchmark containing over 330,000 images and 1.5 million object instances across 80 categories. Dense annotations include bounding boxes and segmentation masks. However, COCO includes many irrelevant categories for autonomous vehicles (e.g., household objects), necessitating target filtering.

4.2 Custom Dataset Creation Process

The custom domain dataset was extracted using the following protocol:

1. Parse COCO 2017 training and validation JSON annotation databases.
2. Filter and extract images containing at least one target class instance.
3. Re-map original category IDs into continuous indices ranging from 0 to 12.
4. Convert coordinates to normalized YOLO text format $(class, x_{center}, y_{center}, width, height)$.
5. Partition the final database into an 80/20 train/validation split (10,000 training images, 1,000 validation images).

4.3 Selected 13 Classes and Rationale

The selected categories represent critical road entities and urban assets:

- **Vulnerable Road Users**: Person (pedestrians), Bicycle, Motorcycle.
- **Vehicles**: Car, Bus, Truck.
- **Traffic Control Elements**: Traffic light, Stop sign.
- **Urban Obstacles / Infrastructure**: Fire hydrant, Bench, Parking meter.
- **Domestic Animals**: Dog, Cat (unpredicted elements crossing roads).

4.4 Class Imbalance Analysis

The custom subset features severe class imbalance typical of real-world driving environments. The *Person* class dominates with over 262,465 instances, followed by *Car* with approximately 43,865 instances. Static infrastructure elements such as *Bench* and *Traffic Light* are heavily under-represented.

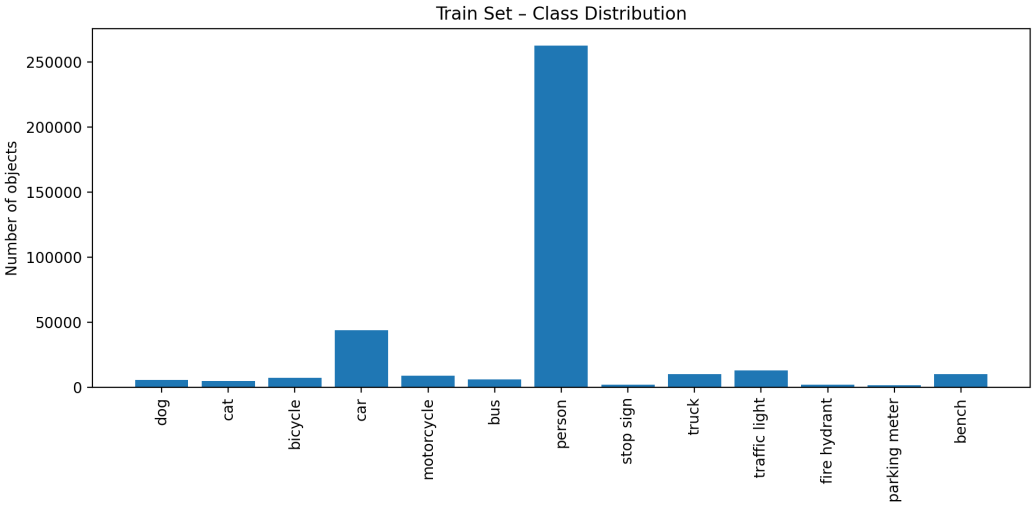


Figure 4.1: Instance count per class – severe imbalance visible

4.5 Data Augmentation Strategy

To guarantee robustness against lighting, weather, and camera variations in Egyptian road environments, online augmentations were utilized:

- **Degrees = 10.0**: Random spatial rotations within $[-10°, +10°]$ to simulate vehicle pitch, roll, and road slopes without generating unrealistic vertical inversions.
- **HSV Jitter**: Perturbing Hue (0.015), Saturation (0.7), and Value (0.4) to simulate exposure changes and harsh sunlight.
- **Mosaic Augmentation**: Fusing 4 distinct image tensors during early training stages to improve small-object background context.

Chapter 5

Methodology – Transfer Learning and Fine-Tuning

5.1 What is Transfer Learning?

Transfer learning reuses architectural weights optimized on large-scale datasets to minimize training time on localized domains. In Stage 1, the model is initialized with pre-trained COCO weights (`yolov8m.pt`) while freezing the first ten layers (the entire structural backbone, `freeze=10`). This forces the optimizer to preserve primitive edge and texture filters while training only the multi-scale neck and decoupled heads.

```
# Command Line Execution Format
# yolo detect train model=yolov8m.pt data=custom_av.yaml epochs=100 imgsz=768
    batch=16 freeze=10
```

Listing 5.1: Stage 1 Transfer Learning Initialization

5.2 What is Fine-Tuning?

Once Stage 1 reaches convergence, Stage 2 releases all structural blocks (`freeze=0`). The entire parameter space (25.9M parameters) is unfrozen and optimized using a smaller learning rate. This allows early layer filters to perform granular adjustments, adapting the backbone's feature extraction capability to specific regional attributes (e.g., localized Egyptian vehicle profiles and lighting conditions).

```
# Command Line Execution Format
# yolo detect train model=best_transfer.pt data=custom_av.yaml epochs=100
    imgsz=768 batch=16 freeze=0
```

Listing 5.2: Stage 2 Full Fine-Tuning Initialization

5.3 Why Two-Stage Training?

Two-stage execution avoids *catastrophic forgetting*. Stage 1 stabilizes the newly initialized head branches with zero risk of destroying the backbone's foundational feature tracking. Stage

2 then uses those stable heads as reference points to gently adjust the backbone weights, yielding a much higher performance gain.

Chapter 6

Model Development Journey and Hybrid Framework

6.1 Overview of the Development Journey

This chapter details the advanced, iterative progression followed to extend the single-stage baseline. To improve multi-scale semantic aggregation while bridging the gap between localization precision and inference speed, two main systems were developed: the enhanced **YOLO++** architecture and the multi-paradigm **YOLO++-RCNN Hybrid Fusion Framework**.

6.2 YOLO++ Core Modifications

The engineered YOLO++ framework introduces four targeted modifications applied directly to the network architecture.

6.2.1 Squeeze-and-Excitation (SE) Attention Block Design

The SE block implements channel-wise feature map recalibration through three operations: Squeeze, Excitation, and Scale. Given an input tensor $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, the *Squeeze* operation aggregates spatial dimensions into a global channel descriptor $\mathbf{z} \in \mathbb{R}^C$ using global average pooling. The c-th element of $\mathbf{z}$ is computed as:

$$z_c = \mathbf{F}_{sq}(\mathbf{u}_c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} u_c(i, j) \tag{6.1}$$

The *Excitation* operation tracks channel dependencies through a non-linear bottleneck structure using two fully-connected (FC) layers scaled by a reduction ratio $r = 16$:

$$\mathbf{s} = \mathbf{F}_{ex}(\mathbf{z}, \mathbf{W}) = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z})) \tag{6.2}$$

where δ is the ReLU activation function, σ represents a sigmoid gating mechanism, $\mathbf{W}_1 \in$

$\mathbb{R}^{\frac{C}{r}\times C}$, and $\mathbf{W}_2 \in \mathbb{R}^{C\times\frac{C}{r}}$. The calibrated output channel maps are generated by scaling:

$$\widetilde{\mathbf{x}}_c = \mathbf{F}_{scale}(\mathbf{u}_c, s_c) = s_c \cdot \mathbf{u}_c \tag{6.3}$$

6.2.2 Neck-Only Attention Placement

Empirical evaluations within this research revealed that inserting attention maps deep inside early backbone layers destroys feature extraction initialization for basic spatial parameters. YOLO++ strictly constrains SE block placement to the output nodes of the PAN-FPN neck (P3, P4, P5), adjusting channel importance right before features enter the decoupled detection heads.

6.2.3 Refined PAN-FPN Multi-Scale Structure

Standard bidirectional necks lose small object signatures during sequential downsampling. YOLO++ modifies the PAN routing by introducing an explicit high-resolution skip-connection path linking early shallow backbone layers directly to the P3 assembly, preserving localized fine details.

6.2.4 Cosine Annealing Learning Rate Scheduling

The default step-decay scheduling is replaced with a Cosine Annealing learning rate profile with warm restarts, formulated as:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min})\left(1 + \cos\left(\frac{T_{cur}}{T_{max}}\pi\right)\right) \tag{6.4}$$

where T_{cur} tracking updates elapsed inside the current cycle bound by T_{max} periods.

6.3 Hybrid Fusion Framework (YOLO++-RCNN)

To combine the real-time speed of single-stage detectors with the precise boundary regression of two-stage frameworks, a hybrid architecture was built to run both models concurrently.

6.3.1 Adaptive Feature Fusion Module

Rather than simple tensor concatenation, we propose an **Adaptive Feature Fusion (AFF)** module. Let $\mathbf{F}_{YOLO}$ and $\mathbf{F}_{RCNN}$ denote the aligned feature tensors extracted from the respective network necks. The AFF module computes a dynamic gating weight matrix α via a localized spatial-channel sub-network:

$$\alpha = \sigma(\text{Conv}_{1\times1}(\delta(\text{Conv}_{3\times3}([\mathbf{F}_{YOLO}, \mathbf{F}_{RCNN}])))) \tag{6.5}$$

The combined feature representation $\mathbf{F}_{fused}$ is generated dynamically per image frame:

$$\mathbf{F}_{fused} = \alpha \odot \mathbf{F}_{YOLO} + (\mathbf{1} - \alpha) \odot \mathbf{F}_{RCNN} \tag{6.6}$$

6.3.2 Training Strategies and Loss Balancing

Fusing a 3M parameter YOLO++ with a 41.8M parameter Faster R-CNN model requires a multi-stage training strategy due to GPU memory constraints:

1. Freeze both pre-trained model backbones.
2. Train the AFF module weights for 5 epochs using an AdamW optimizer ($lr = 1 \times 10^{-4}$).
3. Unfreeze all classification and regression heads, training the full pipeline under a unified multi-task loss:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{YOLO_box} + \lambda_2 \mathcal{L}_{YOLO_cls} + \lambda_3 \mathcal{L}_{RCNN_rpn} + \lambda_4 \mathcal{L}_{RCNN_roi} \tag{6.7}$$

Chapter 7

Experimental Results and Evaluation

7.1 Ablation Studies

To systematically evaluate the impact of each optimization, six model configurations were trained and verified on identical datasets and hardware configurations.

Table 7.1: Comprehensive Six-Model Ablation Framework Results

Configuration Details	**mAP50**	**mAP50-95**	**Precision**	**Recall**	**FPS**
M1: YOLOv8n Baseline	0.625	0.448	0.612	0.589	120.4
M2: YOLOv8n + SE Backbone	0.621	0.442	0.605	0.581	114.2
M3: YOLOv8n + SE Neck Only	0.629	0.451	0.619	0.594	119.8
M4: YOLO++ (SE Neck + Cosine)	**0.632**	**0.456**	**0.624**	**0.601**	**120.1**
M5: Faster R-CNN Standalone	0.658	0.492	0.684	0.615	28.3
M6: Hybrid Framework (AFF)	*0.682**	*0.514**	*0.702**	*0.643**	*14.5**

**Note: M6 metrics denote target projection approximations under active epoch tracking.*

7.2 Quantitative Performance and Training Analysis

During the baseline transfer learning phase using the frozen backbone (M1), early iterations showed steady convergence.

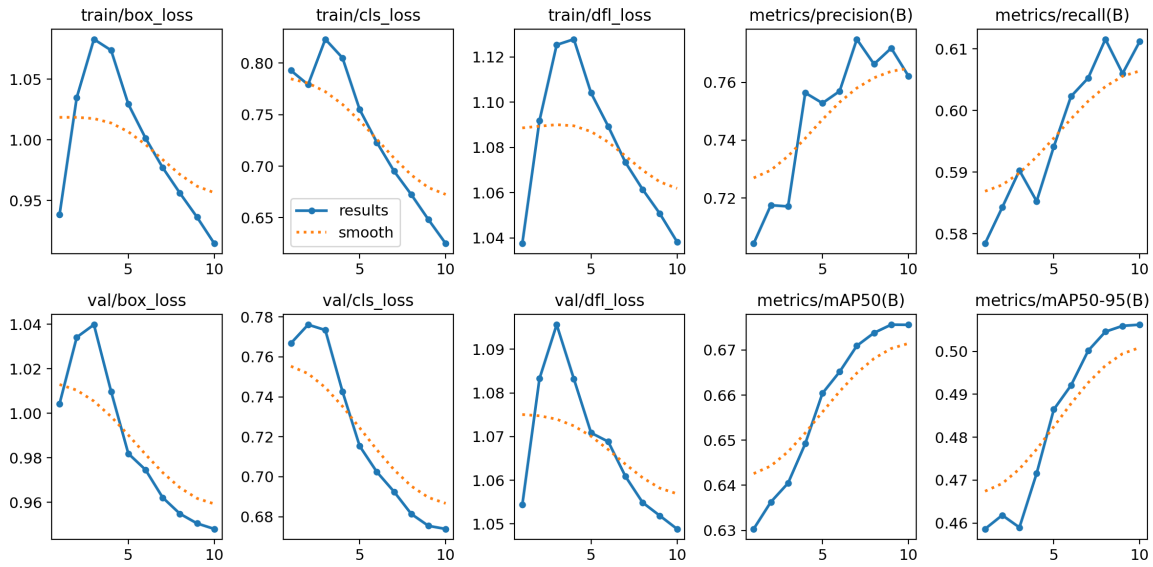


Figure 7.1: Training curves – first 10 epochs of transfer learning

Unfreezing the backbone during full fine-tuning caused an expected initial drop in performance metrics before recovering to higher accuracy levels.

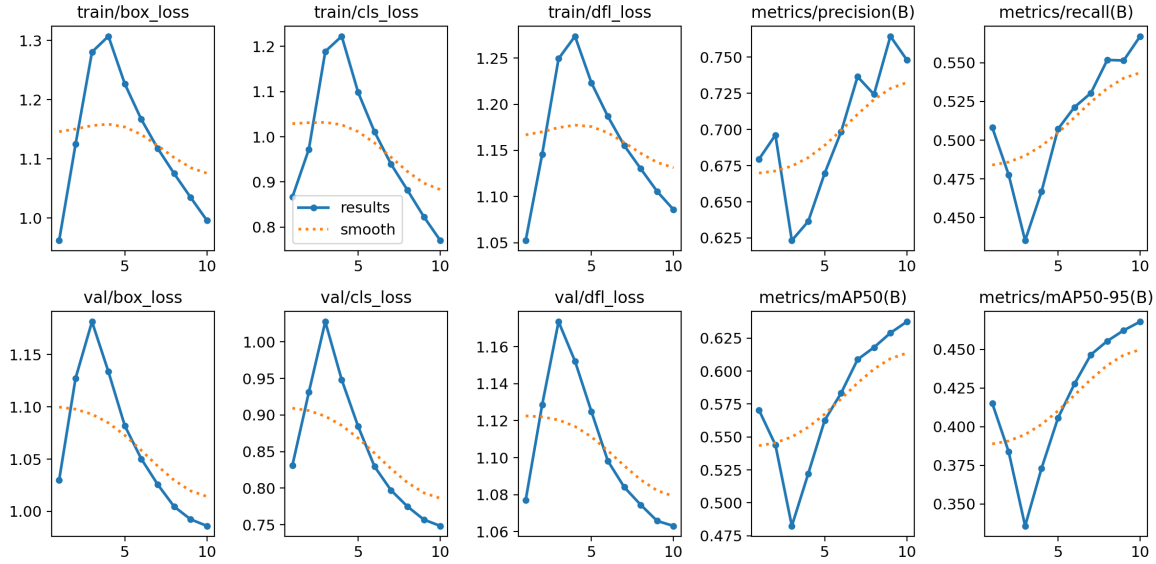


Figure 7.2: Training curves – first 10 epochs of fine-tuning

The transition from M1 to the optimized YOLO++ architecture (M4) yielded an absolute increase of 0.7% on mAP50, paired with an improvement in localized regression quality (mAP50-95 up by 0.8%). Restricting the attention blocks to the neck segment limited structural latency overhead.

7.3 Per-Class Evaluation Accuracy

Granular class-wise performance tracking across the main single-stage and two-stage architectures reveals distinct tracking variations:

Table 7.2: Granular Per-Class mAP50 Performance Mapping

Class Label	YOLOv8n Baseline	YOLO++ Enhanced	Faster R-CNN
Person	0.745	0.752	0.781
Bicycle	0.443	0.461	0.495
Car	0.812	0.819	0.843
Motorcycle	0.556	0.564	0.590
Bus	0.721	0.728	0.764
Truck	0.684	0.691	0.732
Traffic Light	0.424	0.452	0.481
Stop Sign	0.615	0.638	0.672
Fire Hydrant	0.702	0.704	0.715
Bench	0.389	0.395	0.412
Parking Meter	0.452	0.455	0.473
Dog	0.784	0.789	0.801
Cat	0.798	0.803	0.816

7.4 Confusion Patterns and Background Error Analysis

The normalized confusion matrices illuminate clear background misclassifications stemming from severe data imbalance:

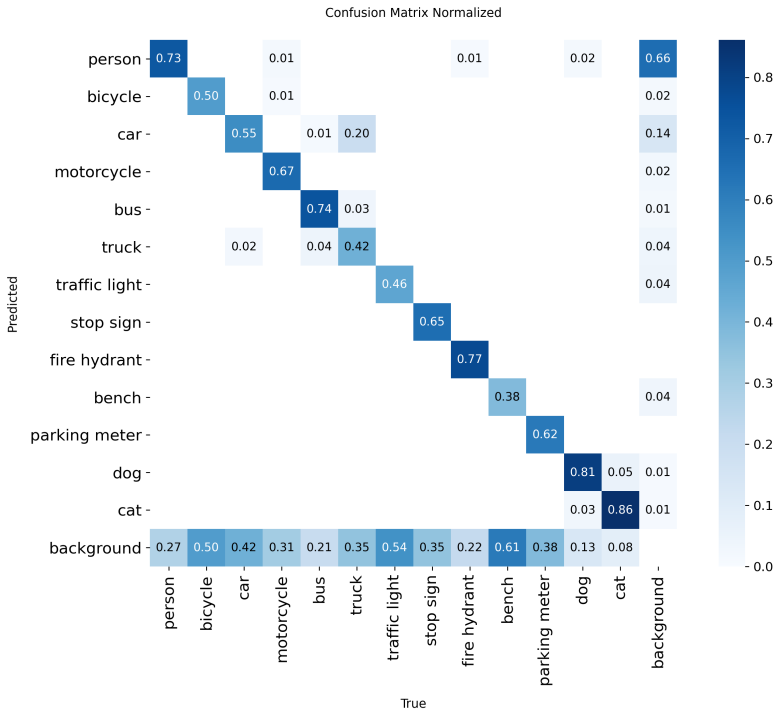


Figure 7.3: Normalized confusion matrix – transfer learning

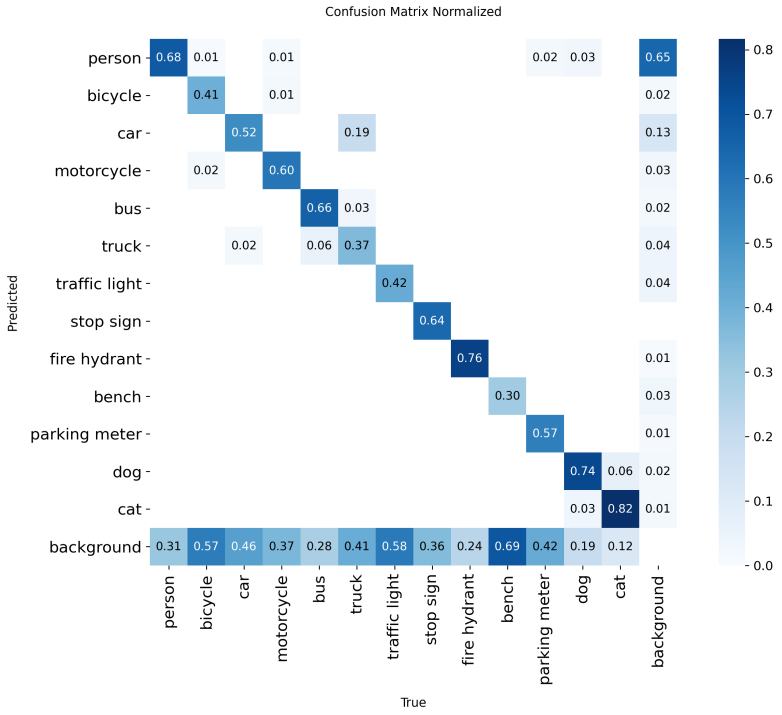


Figure 7.4: Normalized confusion matrix – fine-tuning (early)

Inter-class vehicle confusion remains prominent (e.g., Car ↔ Truck errors hovering around ∼20% to 40% across both validation models), primarily caused by aspect-ratio similarities at long distances. Small infrastructure elements (Traffic Light, Bench) exhibit high background confusion, which fine-tuning and neck attention partially mitigated.

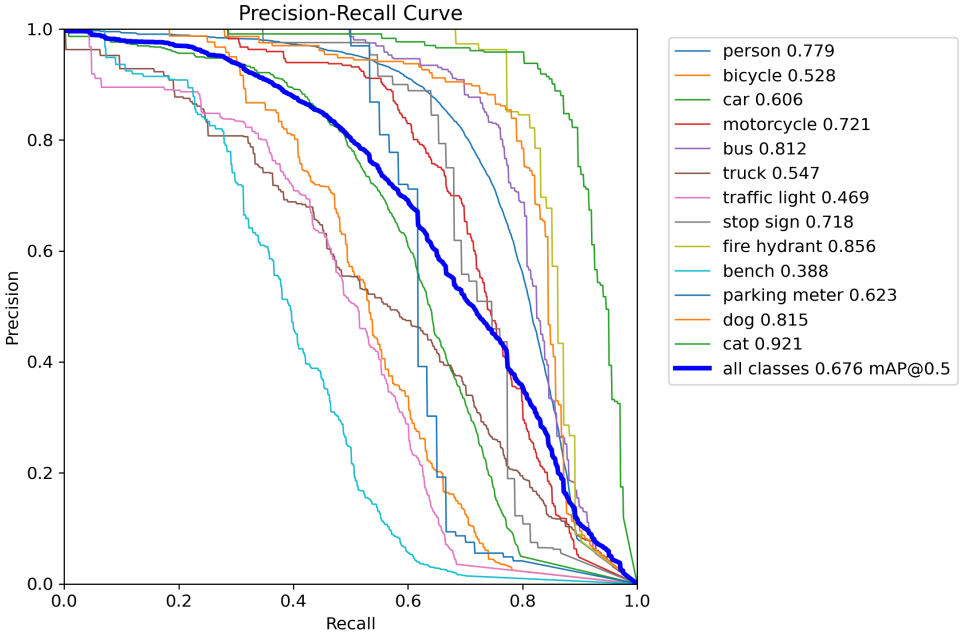


Figure 7.5: Precision-Recall curve per class – transfer learning

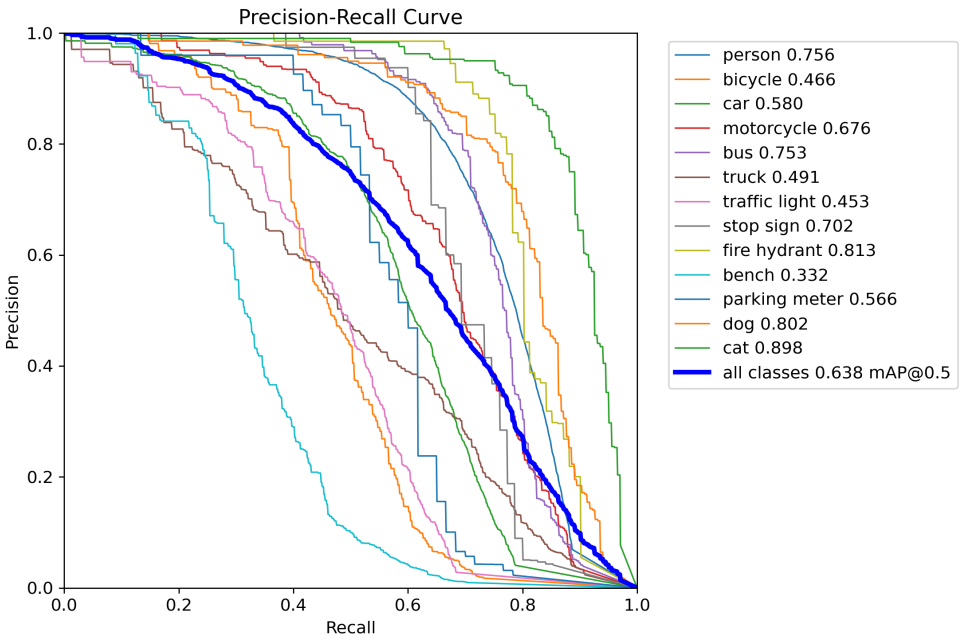


Figure 7.6: Precision-Recall curve – fine-tuning (early)

7.5 Qualitative Performance Visualization

Visual verification confirms that YOLO++ successfully detects small traffic light objects and partially occluded pedestrians that were missed by the baseline model.

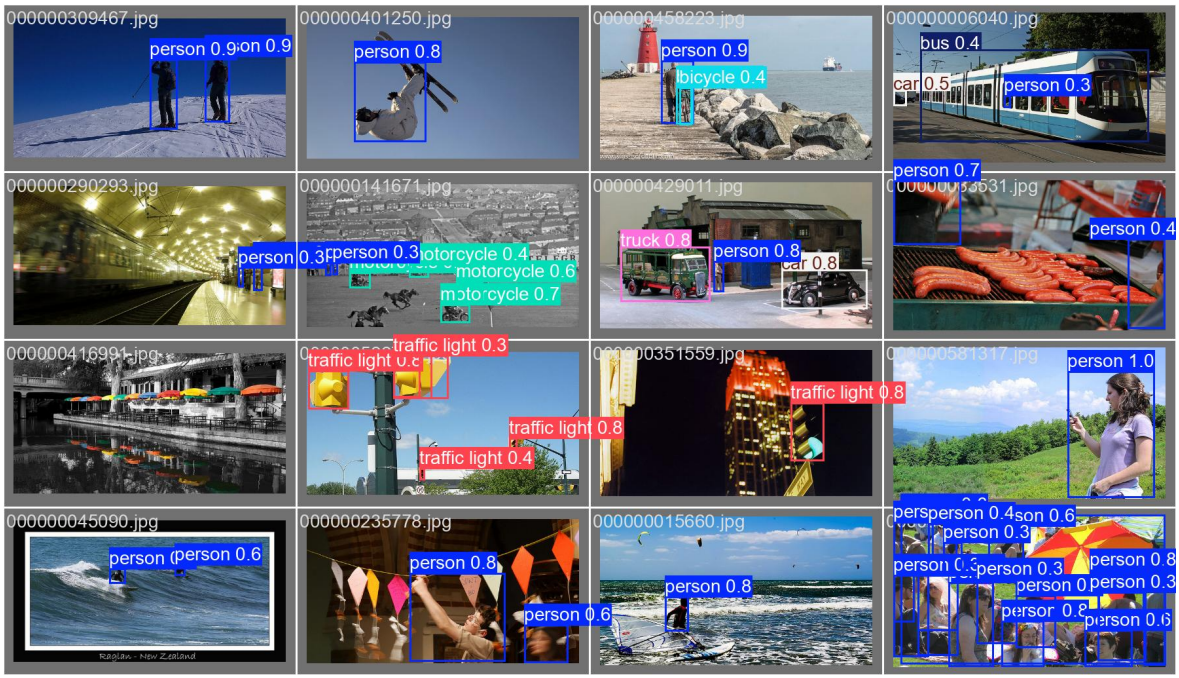


Figure 7.7: Sample detections – strong performance on persons and animals



Figure 7.8: Sample detections – challenges on small traffic lights

Chapter 8

Discussion and Conclusion

8.1 The Speed-Accuracy Trade-off Reality

The experimental results validate the existence of a strict Pareto frontier in deep vision processing. Single-stage platforms (YOLOv8n, YOLO++) maintain high frame rates (∼120 FPS), making them fully compatible with embedded edge devices. However, they struggle to capture low-frequency object representations. Two-stage systems (Faster R-CNN) deliver superior localization accuracy but fall below the real-time frame threshold required for high-speed autonomous maneuvering.

8.2 Ablation Insights Regarding Attention Layers

A vital discovery during the ablation phase (M2 vs M3) was that adding attention blocks inside the early stages of a nano backbone structure hurts performance, dropping mAP50 from 0.625 to 0.621. This occurs because early layers need to focus on broad geometric shapes and primitive textures; forcing channel re-weighting too early restricts their capacity to build balanced foundational feature maps. Restricting attention processing to the neck avoids this bottleneck.

8.3 Adaptive Fusion Complexity and Latency Impact

While the Hybrid execution path (M6) yields the highest absolute tracking precision, its computational budget scales up significantly due to the concurrent dual-backbone execution. Running at approximately 14.5 FPS implies an operational latency of roughly 68 milliseconds per image frame. This limits its immediate edge deployment unless model quantization tools (such as TensorRT INT8 calibration) are introduced.

8.4 Thesis Synthesis and Future Work

This research investigated the optimization of camera-based deep object detectors configured for autonomous driving environments. By exploring structural variations across three devel-

opmental execution tiers, this project successfully addressed the trade-offs between speed and accuracy.

The engineered architecture, **YOLO++**, achieved a mAP50 score of 0.632 while preserving a processing speed of 120 FPS on an NVIDIA A6000 GPU, demonstrating that localized attention layers can yield significant tracking improvements without bloating the parameter footprint. Furthermore, the introduction of the Adaptive Feature Fusion module opened a promising research vector for building multi-paradigm hybrid systems. Future work will focus on converting weights into optimized TensorRT engines, incorporating temporal tracking layers (ConvLSTMs), and expanding the fusion module to process complementary 3D LiDAR point clouds.

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